

TMUA shortcut notebook

10 SHORTCUTS. 22 QUESTIONS. ONE CLEAR REASON.

Recognise the trick.

See it work. Try it yourself.

21 PAST-PAPER QUESTIONS + 1 OFFICIAL SPECIMEN

A short introduction to each trick, followed by one to three examples

 *Before the long algebra...
what would force an answer?*

Start each category with its introduction: when the trick helps, how to use it and what to watch for. Then work through its examples. Every solution shows the first move, the reasoning and the check that makes it valid.

READ THE INK

Blue shows the useful move and circles the answer. Red marks rejected choices and traps. Timings are practice targets once you recognise the method, not measured completion times.

USE IT ACTIVELY

Read the introduction, cover the annotations and attempt the examples. Then uncover the notes and compare your reasoning.

What should I try first?

Choose a category below. Each one begins with a brief introduction, then shows the trick in real TMUA questions.

TRICK / INTRODUCTION	WHEN TO USE IT	EXAMPLES
01 Small n, Big Cut Introduction →	Formula in n ? Try a small allowed value that makes the options disagree.	01 · 02 · 03
02 Shrink the Giant Introduction →	A huge index? Find the repeated step, confirm the rule and count carefully.	04 · 05 · 06
03 Choose Your Own Numbers Introduction →	Free parameters or a “can” claim? Choose simple, valid examples.	07 · 08 · 09
04 Follow the π Introduction →	Circle plus straight edges? Separate the circular area from the piece removed.	10 · 11
05 Mirror, Move, Match Introduction →	Symmetry, rotation or tangency? Use a centre, vertex or axis intersection.	12 · 13 · 14
06 Beat the Options Introduction →	Maximum or parameter range? Find a useful bound or boundary first.	15 · 16 · 17
07 One More Forces It Introduction →	“Guarantee” or “smallest number”? Count the largest avoiding selection; add one.	18
08 Close the Log Loop Introduction →	Logs cycle through the same letters? Multiply around the loop and cancel.	19
09 Cancel Before Calculating Introduction →	Overlapping integrals or a symmetric interval? Use signed areas or oddness.	20 · 21
10 Make the Answers Fight Introduction →	Exactly one true option? Check equivalence and implications between answers.	22

Use the shortest argument that proves enough.

General claim: one failed allowed case eliminates it. If only one listed answer survives, the options can finish the job.

Fixed large index: a small case suggests a pattern. The recurrence or sum must confirm the answer at the requested index.

TRICK 01 · FORMULAS AND CONDITIONS IN N

Small n, Big Cut

When an answer must work for every allowed n , one small value can remove most of the choices.

One failed allowed case is enough to cross an option out.

How to use it

1 Choose a useful small n .

Try 1 or 2. If several choices agree, pick another small value that separates them.

2 Work out the actual answer.

Use the original question with that value of n . Keep all its conditions.

3 Test the options.

Reject every mismatch. If more than one choice remains, test again or finish the remaining reasoning.

Why $n = 2$ can beat $n = 1$

The sum $1 + 2 + \dots + n$ is claimed to be either n^2 or $n(n+1)/2$.

At $n = 1$, both give 1.

At $n = 2$, the sum is 3. The candidates give 4 and 3.

The second test separates the formulas.

Small teaching illustration

Watch for this. A match at one value does not prove a formula. It eliminates the choices that fail; the single-correct-answer format can then finish the job.

NOW SEE IT IN A TMUA QUESTION

01 One ball of each colour →
TMUA early specimen · Paper 1 · Q7

02 A four-term sine test →
TMUA 2020 · Paper 2 · Q15

03 One test removes half the answers →
TMUA 2018 · Paper 2 · Q16

Choose a value that tells the options apart.

One ball of each colour

TMUA early specimen · Paper 1 · Q7 · Official specimen

01

FIRST MOVE Try the smallest legal n that makes the choices disagree.

Target 15–25 s

A bag contains n red, n yellow and n blue balls, where n is a positive integer.

Two balls are drawn at random without replacement. What is the probability that their colours differ?

The quick route✓ Make $n = 1$.

There is just one ball of each colour. Any two drawn must be different. Probability = 1.

✓ Run down the choices.

A, B, D and E become 0. F becomes 1/216. Only C becomes 1.

✓ Legal and decisive.

No replacement matters: you cannot draw the same ball twice.

Answer **C**

~~A~~ $\frac{n-1}{3n-1}$ 0

~~B~~ $\frac{2n-2}{3n-1}$ 0

C $\frac{2n}{3n-1}$ 1 ✓

~~D~~ $\frac{(n-1)^3}{27(3n-1)^3}$ 0

~~E~~ $\frac{3(n-1)}{3n-1}$ 0

~~F~~ $\frac{n^3}{27(3n-1)^3}$ 1/216

WHY THIS SHORTCUT IS VALID

Every offered formula claims to work for all positive integers n . Five fail at $n = 1$. A direct check is $2n/(3n - 1)$: after the first draw, $2n$ of the remaining $3n - 1$ balls have a different colour.

One tiny bag replaces the probability algebra.

A four-term sine test

TMUA 2020 · Paper 2 · Q15

02

FIRST MOVE Try the smallest legal n that makes the choices disagree.

Target 25–40 s

For positive integer n , which condition is necessary and sufficient for

$$\sum_{k=1}^n \sin \frac{k\pi}{3} = \frac{\sqrt{3}}{2}?$$

~~A~~ $n = 1$ misses 4

~~B~~ n is a multiple of 3. misses 4

~~C~~ n is a multiple of 6. misses 4

D n is 1 more than a multiple of 3. accepts 4

~~E~~ n is 1 more than a multiple of 6. misses 4

~~F~~ n is 1 or 2 more than a multiple of 6. misses 4

The quick route

✓ $n = 1$ leaves a crowd.

A, D, E and F all accept 1. So choose a value that splits them.

✓ Try $n = 4$.

Write $s = \sqrt{3}/2$.

The sum is

$$s + s + 0 - s = s.$$

Only D accepts $n = 4$.

✓ Quick pattern check.

Each six-term block sums to 0. Within a block, the target occurs after terms 1 and 4. That is $n \equiv 1 \pmod{3}$.

Answer **D**

WHY THIS SHORTCUT IS VALID

The valid value $n = 4$ eliminates A, B, C, E and F because a necessary condition must accept every solution. The six-term sine cycle confirms the surviving condition for all n .

Choose the small value that separates, not merely the first value.

One test removes half the answers

TMUA 2018 · Paper 2 · Q16

03**FIRST MOVE** Try the smallest legal n that makes the choices disagree.

Target 40–60 s

An arithmetic progression $x_1, x_2, \dots$ consists of integers. The median of its first n terms is an integer.

Which medians must also be integers?

- I. The first $n + 2$ terms.
- II. The first $2n$ terms.
- III. $x_2, x_4, \dots, x_{2n}$.

- A None
- B I only
- ~~C~~ II only *II fails*
- D III only
- ~~E~~ I and II *II fails*
- F** I and III
- ~~G~~ II and III *II fails*
- ~~H~~ I, II and III *II fails*

Eliminate, then finish

✓ Use 1, 2, 3, ... and $n = 1$.

*The given median is 1.
But the first $2n$ terms
are 1, 2, with median $1\frac{1}{2}$.
II is not forced.*

✓ Cross out every option
with II.

*Remove C, E, G and H.
A, B, D and F remain.
Write the first term as a
and the common
difference as d .*

✓ Check statements I and
III.

*I: new median = old
median + d .
III: the median is $a + nd$.
Both are integers.
I + III → F*

*Answer **F***

WHY THIS SHORTCUT IS VALID

For any arithmetic progression, the median is the mean of its first and last terms, even when $d < 0$. The first $n + 2$ median equals the first n median plus d ; the even-indexed subsequence has median $a + nd$.

A counterexample clears half the choices; two short checks finish.

TRICK 02 · LARGE INDICES, RECURRENCES AND SUMS

Shrink the Giant

A huge index often means repeating one simple step. Look for that step before doing any long calculation.

Find the small rule hiding inside the big number.

How to use it

1 Look at one or two steps.

Does each term double? Does an exponent halve? Does each step add another power?

2 State the rule.

Use the recurrence or the structure of the sum. A few matching values only suggest a pattern.

3 Count from the correct start.

Term 1 → term 100: 99 steps. Term 0 → term 100: 100 steps.

Optional: shrink the index

If the question uses 100 and the options contain 99, replace them consistently by n and $n - 1$.

Try $n = 1$ or 2 to spot an indexing error or a likely formula.

If the rule is already obvious, use it straight away.

Small teaching illustration

Watch for this. Changing 100 to 2 changes the problem. A small-case match does not prove the answer at 100. Confirm the recurrence or sum.

NOW SEE IT IN A TMUA QUESTION

04

The hundredth term

TMUA 2019 · Paper 1 · Q4



05

A thousand integrals?

TMUA 2017 · Paper 2 · Q6



06

Ninety-nine steps on a staircase

TMUA 2023 · Paper 2 · Q17



Small cases suggest; the rule confirms.

The hundredth term

TMUA 2019 · Paper 1 · Q4

04

FIRST MOVE Find what one step does; use the rule to handle the large index.

Target 15–25 s

A sequence satisfies

$$x_1 = 10, \quad x_{n+1} = \sqrt{x_n}.$$

Which expression equals x_{100} ?~~A~~ $10^{2^{99}}$ grows~~B~~ $10^{2^{100}}$ grows**C** $10^{2^{-99}}$ 99 halvings~~D~~ $10^{2^{-100}}$ 100 halvings~~E~~ $10^{-2^{99}}$ below 1~~F~~ $10^{-2^{100}}$ below 1~~G~~ $10^{-2^{-99}}$ below 1~~H~~ $10^{-2^{-100}}$ below 1

The quick route

✓ Watch the exponent.Write $x_n = 10$ raised to a power.

A square root halves that power.

It starts at 1.

✓ Count the steps carefully.From x_1 to x_{100} there are 99 steps.After 99 halvings, the exponent is 2^{-99} .✓ Match the answer.**C**~~D~~ uses 100 halvings: one too many.

The initial term is not a square-root step.

Answer **C**

WHY THIS SHORTCUT IS VALID

Writing $x_n = 10^{e_n}$ gives $e_1 = 1$ and $e_{n+1} = e_n/2$, so $e_n = 2^{-(n-1)}$. Small cases can check the indexing; this rule proves the answer for the requested term.

A large index becomes a count of identical small steps.

A thousand integrals?

TMUA 2017 · Paper 2 · Q6

05

FIRST MOVE Find what one step does; use the rule to handle the large index.

Target 15–25 s

Let $u_0 = 1$, and for integers $n \geq 1$ define

$$u_n = \int_0^1 4x u_{n-1} dx.$$

Find u_{1000} .

- A** 2^{1000} 1000 doublings
- ~~B~~ 4^{1000} wrong multiplier
- ~~C~~ $\frac{4}{1000!}$ no factorial
- ~~D~~ $\frac{4}{1001!}$ no factorial
- ~~E~~ $\frac{2^{1000}}{1000!}$ no factorial
- ~~F~~ $\frac{4^{1000}}{1000!}$ no factorial
- ~~G~~ $\frac{2^{1000}}{1001!}$ no factorial
- ~~H~~ $\frac{4^{1000}}{1001!}$ no factorial

The quick route

✓ Pull out the constant.

u_{n-1} is a number, so it stays constant while x is integrated.

✓ Each step doubles.

$$\int_0^1 4x dx = 2.$$

Therefore

$$u_n = 2u_{n-1}.$$

✓ Start at index zero.

$$u_0 = 1.$$

After 1000 doublings:

$$u_{1000} = 2^{1000} \rightarrow \text{A.}$$

Answer **A**

WHY THIS SHORTCUT IS VALID

$u_n = u_{n-1} \int_0^1 4x dx = 2u_{n-1}$. Since $u_0 = 1$, the sequence is $u_n = 2^n$. The multiplier is already clear, so testing eight options at a smaller index would add unnecessary work.

Find the multiplier once; apply it to every step.

Ninety-nine steps on a staircase

TMUA 2023 · Paper 2 · Q17

06

FIRST MOVE Find what one step does; use the rule to handle the large index.

Target 25–40 s

Write $\lceil x \rceil$ for the least integer greater than or equal to x .

Evaluate

$$\int_0^{99} 2^{\lceil x \rceil} dx.$$

~~A~~ 2^{99} 4

~~B~~ $2^{99} - 1$ 3

~~C~~ $2^{99} - 2$ 2

~~D~~ 2^{100} 8

~~E~~ $2^{100} - 1$ 7

F $2^{100} - 2$ 6 ✓

Test, then confirm

✓ Shrink the upper limit to 2.

The two unit strips have heights 2 and 4.

Small-case area = 6.

✓ Shrink every option consistently.Replace 99 by n , 100 by $n + 1$, then set $n = 2$.

The options give 4, 3, 2, 8, 7, 6.

F is the candidate.

✓ Confirm by cancellation.

$$S = 2 + 4 + \dots + 2^{99}.$$

$$2S - S = 2^{100} - 2.$$

F

—

Answer **(F)**

WHY THIS SHORTCUT IS VALID

On each interval $(k-1, k]$, the integrand is 2^k . Endpoint values do not affect area. Thus the integral is the geometric sum $\sum_{k=1}^{99} 2^k = 2^{100} - 2$.

A two-step staircase exposes the endpoint correction.

TRICK 03 · FREE PARAMETERS AND “POSSIBLE” CLAIMS

Choose Your Own Numbers

When a proposed answer must work generally, replace awkward letters with convenient values. When the question asks what is possible, build an example.

Choose easy numbers that still separate the answers.

How to use it

1 Keep the conditions.

If a parameter must be positive, do not set it to zero. Keep denominators nonzero.

2 Make the arithmetic simple.

Choose values that give an easy graph, interval, probability or count.

3 Know what the example proves.

One failed case disproves “must”. One successful example proves “can”.

Avoid a useless coincidence

A square of side a has area a^2 . Compare the candidate formulas a^2 and $2a$.

At $a = 2$, both give 4.

At $a = 3$, the area is 9; $2a$ gives 6 and fails.

“Easy” is useful only when it tells the choices apart.

Small teaching illustration

Watch for this. Do not impose a new assumption on a fixed problem. Choosing a convenient example is valid only when the proposed claim must cover that example.

NOW SEE IT IN A TMUA QUESTION

07 Two quadratics, six intervals

TMUA 2017 · Paper 1 · Q19



08 Shift the minimum to an easy point

TMUA 2023 · Paper 1 · Q11



09 “Possible” asks for examples

TMUA 2023 · Paper 2 · Q19



For “must”, try to break it. For “can”, construct it.

Two quadratics, six intervals

TMUA 2017 · Paper 1 · Q19

07

FIRST MOVE Choose simple legal values that separate the answers.

Target 25–40 s

The complete solution of $x^2 + bx + c < 0$ is $p < x < q$,
where b, c, p, q are real and $c < 0$.

Which interval solves

$$x^2 + bcx + c^3 < 0?$$

~~A~~ $\frac{p}{c} < x < \frac{q}{c}$ *reversed*

~~B~~ $\frac{q}{c} < x < \frac{p}{c}$ *too narrow*

~~C~~ $pc < x < qc$ *reversed*

D $qc < x < pc$ *(-8, 8)*

~~E~~ $pc^2 < x < qc^2$ *too wide*

~~F~~ $qc^2 < x < pc^2$ *reversed*

The quick route

✓ Choose $b = 0, c = -4$.

Then $p = -2$ and $q = 2$.

The new inequality is
 $x^2 - 64 < 0$.

✓ Match the interval.

We need $-8 < x < 8$.

Only D gives that.

B is $(-\frac{1}{2}, \frac{1}{2})$; E is $(-32, 32)$.

✓ Do not make every option agree.

$c = -1$ makes B, D and E agree in this example.

Choose -4 , with a larger absolute value, to separate them.

Answer **D**

WHY THIS SHORTCUT IS VALID

Each option proposes a general answer, so one legal counterexample eliminates it. Structurally, the new polynomial is $(x - cp)(x - cq)$; since $c < 0$, the endpoint order reverses.

Choose numbers that separate the answers.

Shift the minimum to an easy point

TMUA 2023 · Paper 1 · Q11

08

FIRST MOVE Choose simple legal values that separate the answers.

Target 25–40 s

Let $f(x) = x^2 - 6x$. The graphs of $y = f(kx)$ and $y = f(x - c)$, with $k, c > 0$, have their minimum at the same point.

Which expression gives k in terms of c ?

~~A~~ $\frac{3-c}{3}$ 0

B $\frac{3}{c+3}$ 1/2 ✓

~~C~~ $\frac{c-6}{6}$ -1/2

~~D~~ $\frac{6}{6-c}$ 2

~~E~~ $\frac{c+9}{9}$ 4/3

~~F~~ $\frac{9}{c-9}$ -3/2

The quick route

✓ Choose $c = 3$.

f has its minimum at input 3.

$f(x - 3)$ reaches that input when $x = 6$.

✓ The other graph must match.

For $f(kx)$, the minimum occurs when $kx = 3$.

At $x = 6$ this gives $k = \frac{1}{2}$.

✓ Plug $c = 3$ into the choices.

They give 0, $\frac{1}{2}$, $-\frac{1}{2}$, 2, $\frac{4}{3}$, $-\frac{3}{2}$.

Only B works.

Answer **(B)**

WHY THIS SHORTCUT IS VALID

Both transformations preserve the minimum value -9 . Their minimum locations are $3/k$ and $3 + c$. The special case eliminates five formulas; equating locations confirms $k = 3/(c + 3)$.

A convenient shift replaces symbolic transformation algebra.

“Possible” asks for examples

TMUA 2023 · Paper 2 · Q19

09

FIRST MOVE Choose simple legal values that separate the answers.

Target 25–40 s

For a non-constant real polynomial f , put $g(x) = xf'(x)$. Let M and N be the numbers of distinct real zeros of f and g , respectively.

Which situations are possible?

- I. $M < N$
 - II. $M = N$
 - III. $M > N$
- A None
 - B I only
 - C II only
 - D III only
 - E I and II
 - F I and III
 - G II and III
 - H** I, II and III

The quick route

✓ Keep g fixed; move f .

Try $f(x) = x^2 + c$.

Then $g(x) = 2x^2$, so $N = 1$ for every c .

✓ Three small choices.

$c = 1$: $M = 0 \rightarrow \text{I.}$

$c = 0$: $M = 1 \rightarrow \text{II.}$

$c = -1$: $M = 2 \rightarrow \text{III.}$

All three $\rightarrow \text{H.}$

✓ Distinct means different.

x^2 has one real zero, even though that zero has multiplicity two.

Answer **H**

WHY THIS SHORTCUT IS VALID

Each statement says a situation can occur. One valid polynomial for each situation proves possibility. No general theorem comparing all polynomial root counts is needed.

For “can”, construct; for “must”, try to break.

TRICK 04 · CIRCULAR AREAS AND STRAIGHT-EDGED PIECES

Follow the π

A curved region often becomes simple once you separate the circular area from the triangles, rectangles or trapezia.

Circular contribution minus the area you remove.

How to use it

1 Identify the circular piece.

Is it a full circle, a semicircle or a known sector? Find its exact area.

2 Identify what is removed.

Use the actual triangle, rectangle or trapezium areas. Keep the coefficients clear.

3 Compare complete expressions.

Use the structure to remove options, then check the remaining subtraction.

A segment in one line

Take a quarter-circle of radius 2. Remove the triangle joining its centre to the two ends of the arc.

$$\text{sector} = \frac{1}{4}\pi(2^2) = \pi$$

$$\text{triangle} = \frac{1}{2}(2)(2) = 2$$

The remaining segment has area $\pi - 2$.

Small teaching illustration

Watch for this. A circle does not guarantee that π survives in the final answer: circular terms can cancel. Its irrationality alone does not identify an option.

NOW SEE IT IN A TMUA QUESTION

10 **Keep the semicircle; subtract the strips**

TMUA 2023 · Paper 1 · Q10



11 **A forbidden disk in the centre map**

TMUA 2022 · Paper 1 · Q19



Get the π term from the geometry, not from a guess.

Keep the semicircle; subtract the strips

TMUA 2023 · Paper 1 · Q10

10

FIRST MOVE Identify the circular contribution and the piece being subtracted.

Target 30–50 s

The trapezium rule, using four equal strips, approximates

$$\int_{-2}^2 \sqrt{4-x^2} dx.$$

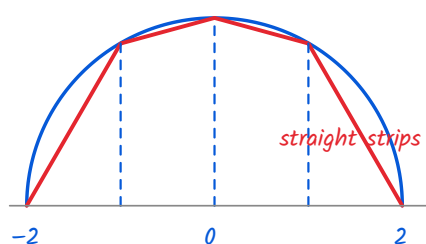
What is the positive difference between the exact value and the approximation?

The quick route✓ See the semicircle.Radius 2 gives exact area $\frac{1}{2}\pi \times 2^2 = 2\pi$.

The four strips underestimate the curved area.

✓ The straight strips have no π .Their heights are 0, $\sqrt{3}$, 2, $\sqrt{3}$, 0.Width 1 gives $T = 2 + 2\sqrt{3}$.✓ Subtract once.

$$\begin{aligned} \text{Error} &= 2\pi - 2 - 2\sqrt{3} \\ &= 2(\pi - 1 - \sqrt{3}). \end{aligned}$$

BAnswer **(B)**~~A~~ $2(\pi - 2 - 2\sqrt{3})$ negative**B** $2(\pi - 1 - \sqrt{3})$ $2\pi - T$ ~~C~~ $2(2\pi - 1 - \sqrt{3})$ 4π term~~D~~ $4(\pi - 1 - \sqrt{3})$ 4π term~~E~~ $2\pi - 3\sqrt{3}$ wrong strips~~F~~ $4\pi - 6\sqrt{3}$ 4π term

Teaching sketch · not to scale

WHY THIS SHORTCUT IS VALID

The geometry supplies the exact circular term; the trapezium calculation supplies the rest. “There is a circle” or “ π is irrational” alone would not justify selecting an answer. Here the decomposition explicitly gives $2\pi - (2 + 2\sqrt{3})$.

Track the circular contribution, then check the subtraction.

A forbidden disk in the centre map

TMUA 2022 · Paper 1 · Q19

11

FIRST MOVE Identify the circular contribution and the piece being subtracted.

Target 35–55 s

Circle C_1 has equation $x^2 + y^2 = 25$. Circle C_2 has radius 4 and centre (a, b) .

The centre is chosen uniformly from the rectangle $-2 \leq a \leq 2$, $-3 \leq b \leq 3$.

What is the probability that the circles intersect?

The quick route

✓ Move the centre, not the circles.

Two circles of radii 5 and 4 meet when centre distance d lies between 1 and 9.

✓ The outer limit is automatic.

Here $d \leq \sqrt{13} < 9$.
Failure occurs only inside the unit disk $d < 1$.

✓ Rectangle minus disk.

All centre positions have area $4 \times 6 = 24$.

Forbidden area = π .

$$P = (24 - \pi)/24 \rightarrow F$$

Answer **(F)**

~~A~~ $\frac{1}{25}$ misses disk

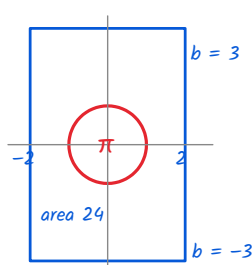
~~B~~ $\frac{9}{25}$ misses disk

~~C~~ $\frac{16}{25}$ misses disk

~~D~~ $\frac{6-\pi}{6}$ wrong total

~~E~~ $\frac{16-\pi}{24}$ wrong total

(F) $\frac{24-\pi}{24}$



Teaching sketch · not to scale

WHY THIS SHORTCUT IS VALID

The unit disk lies wholly inside the centre rectangle. Tangency boundaries have zero area, so including or excluding them does not change the probability. The denominator is the area of possible centre positions, not the area of either original circle.

Turn random geometry into an area on the centre map.

TRICK 05 · SYMMETRY, TRANSFORMATIONS AND TANGENCY

Mirror, Move, Match

Many geometry questions can be decided by one centre, one vertex or one intersection with an axis.

Track the feature that distinguishes the options.

How to use it

1 Bisect an area: mark the centre.

A line through the centre bisects a circle or square. Join distinct centres to bisect both shapes.

2 Rotate: move one landmark.

Track the vertex relative to the rotation centre. Check which way the curve opens.

3 For tangency: use a double root.

Use $x = 0$ for the y -axis. One repeated root means the quadratic discriminant is zero.

One point through a turn

A point 3 units right of a rotation centre moves 3 units below it after a 90° clockwise turn.

Track that displacement first; the full coordinate transformation may be unnecessary.

One landmark can identify the whole offered curve.

Small teaching illustration

Watch for this. Do not assume symmetry unless the shape has it. A reflection preserves tangency, but can reverse the signs of the centre coordinates.

NOW SEE IT IN A TMUA QUESTION

12 A line through two centres

TMUA 2021 · Paper 2 · Q7



13 Touch the axis once

TMUA 2021 · Paper 2 · Q15



14 Rotate just the vertex

TMUA 2021 · Paper 1 · Q18



Centre, vertex or axis: pick the smallest useful feature.

A line through two centres

TMUA 2021 · Paper 2 · Q7

12

FIRST MOVE Use a centre, a vertex or an axis intersection.

Target 20–35 s

A line divides each of these shapes into two equal areas:

- the circle $(x - 9)^2 + (y + 2)^2 = 4$;
- the square with vertices $(1, 0)$, $(1, 2)$, $(-1, 2)$, $(-1, 0)$.

Where does the line meet the x -axis?*The quick route*✓ Bisect through the centres.

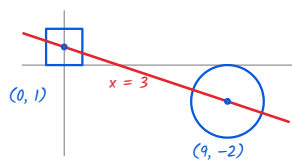
Square centre: $(0, 1)$.
 Circle centre: $(9, -2)$.
 The required line joins them.

✓ Read the small slope.

Change in $y = -3$;
 change in $x = 9$.
 So $y = 1 - x/3$.

✓ Meet the x -axis.

Set $y = 0$.
 $x = 3 \rightarrow \mathbf{B}$
No area integrals needed.

Answer **(B)**

Teaching sketch · not to scale

WHY THIS SHORTCUT IS VALID

For these filled, centrally symmetric convex shapes, the line through the centre in any given direction bisects the area, and an offset parallel line cannot also bisect it. The distinct centres therefore determine the unique common area-bisecting line.

Two area conditions become two points.

Touch the axis once

TMUA 2021 · Paper 2 · Q15

13

FIRST MOVE Use a centre, a vertex or an axis intersection.

Target 15–25 s

A circle has equation

$$x^2 + ax + y^2 + by + c = 0,$$

where $a, b, c \neq 0$. Which condition is necessary and sufficient for the circle to be tangent to the y -axis?

To shorten the options, write its radius as

$$R = \sqrt{\frac{a^2 + b^2}{4}} - c.$$

~~A~~ $a^2 = 4c$ *wrong axis*

B $b^2 = 4c$

~~C~~ $\frac{a}{2} = R$ *one side only*

~~D~~ $\frac{b}{2} = R$ *wrong axis*

~~E~~ $-\frac{a}{2} = R$ *one side only*

~~F~~ $-\frac{b}{2} = R$ *wrong axis*

The quick route

✓ The y -axis means $x = 0$.

Substitute $x = 0$ into the circle equation:

$$y^2 + by + c = 0.$$

✓ Tangency means one repeated root.

A quadratic has a repeated root when its discriminant is zero:

$$b^2 - 4c = 0.$$

$$b^2 = 4c \rightarrow \text{B.}$$

✓ Keep the axes straight.

Setting $y = 0$ would test tangency to the x -axis instead.

Answer **B**

WHY THIS SHORTCUT IS VALID

The roots of $y^2 + by + c = 0$ are the intersections with the y -axis. Tangency gives one repeated root, hence $b^2 - 4c = 0$. Equivalently, the distance condition is $|a|/2 = R$, which covers circles on either side of the axis; C and E each miss one side.

Set the coordinate to zero, then ask for a repeated root.

Rotate just the vertex

TMUA 2021 · Paper 1 · Q18

14

FIRST MOVE Use a centre, a vertex or an axis intersection.

Target 30–45 s

The curve

$$x = y^2 - 6y + 11$$

is rotated through 90° clockwise about $(-2, 3)$.

Which equation describes the new curve?

~~A~~ $y = -x^2 - 4x - 3$ vertex too high

B $y = -x^2 - 4x - 5$

~~C~~ $y = -x^2 - 6x - 7$ vertex $x = -3$

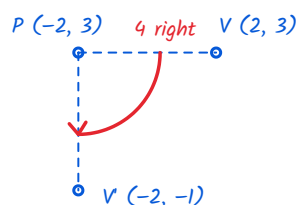
~~D~~ $y = -x^2 - 6x - 11$ vertex $x = -3$

~~E~~ $y = x^2 - 4x + 5$ opens up

~~F~~ $y = x^2 + 4x + 3$ opens up

~~G~~ $y = x^2 - 6x + 11$ opens up

~~H~~ $y = x^2 + 6x + 7$ opens up



The quick route

✓ Locate just the vertex.

$$x = (y - 3)^2 + 2.$$

Vertex $(2, 3)$ is 4 units right of the rotation centre.

✓ Right turns into down.

The new vertex is $(-2, -1)$.

The curve opens downwards, so E–H disappear.

✓ Match the remaining vertices.

A and B have x-coordinate -2 . A gives height 1; B gives -1 .

B

Answer **(B)**

WHY THIS SHORTCUT IS VALID

A rigid rotation preserves the parabola's shape. C and D have vertex x-coordinate -3 , so they cannot work. The transformed equation is $y = -(x + 2)^2 - 1$.

One landmark and one direction replace a full substitution.

TRICK 06 · MAXIMA, MINIMA AND PARAMETER RANGES

Beat the Options

You may be able to beat the distractors without calculating the exact optimum. For a parameter range, an easy boundary can reveal the answer.

A useful bound can be enough to choose the answer.

How to use it

1 For a maximum: build one valid case.

The maximum cannot be below this value. Eliminate every option that is smaller.

2 For a range: test a boundary.

Try where an absolute value is zero, or a square root is smallest.

3 Check what still needs proving.

If several choices remain, continue. For a complete range, show that every surviving value works.

You already beat two choices

Suppose you can construct a valid shape with area 12.

If the offered maximum areas are 8, 10 and 13, the first two are impossible.

The maximum must be at least 12; only 13 remains.

Small teaching illustration

Watch for this. A trial shape need not be optimal. It only needs to eliminate the other options. A necessary parameter bound alone does not prove sufficiency.

NOW SEE IT IN A TMUA QUESTION

15 A rectangle that beats seven options
TMUA 2022 · Paper 1 · Q15



16 Hit the cusp, then prove the boundary
TMUA 2019 · Paper 2 · Q18



17 Ask whether the target is reachable
TMUA 2018 · Paper 2 · Q20



First ask: how much do I really need to calculate?

A rectangle that beats seven options

TMUA 2022 · Paper 1 · Q15

15

FIRST MOVE Find an easy bound or boundary, then check what it proves.

Target 25–40 s

A rectangle lies in the region enclosed by $y = 8 - 2x^2$ and $y = x^2 - 2$. Its sides are parallel to the coordinate axes.

What is its greatest possible area?

The quick route

✓ Build one easy rectangle.

Choose vertical sides $x = -1$ and $x = 1$.

Top height = 6.

Bottom height = -1 .

✓ Its area is already 14.

Width $2 \times$ height $7 = 14$.

So the maximum must be at least 14.

✓ Seven answers are too small.

A–G are all below 14.

Only H can be the maximum.

H — no differentiation.

Answer **(H)**

~~A~~ $\frac{26}{9} < 14$

~~B~~ $\frac{52}{9} < 14$

~~C~~ $\frac{4\sqrt{6}}{3} < 14$

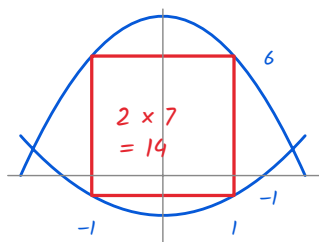
~~D~~ $\frac{8\sqrt{6}}{3} < 14$

~~E~~ $4\sqrt{2} < 14$

~~F~~ $8\sqrt{2} < 14$

~~G~~ $\frac{20\sqrt{10}}{9} < 14$

(H) $\frac{40\sqrt{10}}{9}$ only survivor



Teaching sketch · not to scale

WHY THIS SHORTCUT IS VALID

This is a valid inscribed rectangle. For quick exact comparisons, $\sqrt{2} < 3/2$, $\sqrt{6} < 3$, and $\sqrt{10} < 4$ put every option A–G below 14. Under the single-correct-answer format, H is forced; the trial rectangle itself need not be optimal.

You do not need the maximum if one feasible case beats the distractors.

Hit the cusp, then prove the boundary

TMUA 2019 · Paper 2 · Q18

16

FIRST MOVE Find an easy bound or boundary, then check what it proves.

Target 35–55 s

For which complete set of real values of a does

$$a|x| + 1 \leq |x - 2|$$

hold for every real x ?

Eliminate, then finish

✓ Make the right side vanish.At $x = 2$:

$$2a + 1 \leq 0.$$

So $a \leq -\frac{1}{2}$ is necessary.✓ Check $a = -\frac{1}{2}$ really works.

Triangle inequality gives

$$|x| + |x - 2| \geq 2.$$

$$\text{Therefore } \frac{1}{2}|x| + |x - 2| \geq 1.$$

✓ Smaller a only helps.Decreasing a lowers the left side for every x .

$$a \leq -\frac{1}{2} \rightarrow E$$

Answer (E)

~~A~~ $a \leq \frac{3}{2}$ too broad

~~B~~ $a \leq 1$ too broad

~~C~~ $a \leq \frac{1}{2}$ too broad

~~D~~ $a \leq 0$ too broad

E $a \leq -\frac{1}{2}$

~~F~~ $a \leq -1$ too narrow

~~G~~ $a \leq -\frac{3}{2}$ too narrow

~~H~~ No value of a works. boundary works

WHY THIS SHORTCUT IS VALID

The sufficiency step is $\frac{1}{2}|x| + |x - 2| \geq \frac{1}{2}(|x| + |x - 2|) \geq 1$. This verifies the boundary for all real x , and monotonicity in a gives the entire range. The test $x = 2$ alone would not establish completeness.

A sharp-point test gives the boundary; one inequality confirms it.

Ask whether the target is reachable

TMUA 2018 · Paper 2 · Q20

17

FIRST MOVE Find an easy bound or boundary, then check what it proves.

Target 30–45 s

For which complete set of real p does the equation

$$\sqrt{x+p} + \sqrt{x} = p$$

have at least one real solution x ?

The quick route

✓ Start with the domain.

The left side is non-negative, so $p \geq 0$. Also $x \geq 0$.

✓ Find the lowest left side.

For $p \geq 0$, it starts at $\sqrt{p}$ when $x = 0$, then increases without bound. The target p is reachable iff $p \geq \sqrt{p}$.

✓ Split off $p = 0$.

$p = 0$ works.
For $p > 0$, divide $p \geq \sqrt{p}$ by $\sqrt{p}$ to get $\sqrt{p} \geq 1$.
 $p = 0$ or $p \geq 1 \rightarrow \mathbf{B}$.

Answer **(B)**~~A~~ $p = 0$ or $p = 1$ misses $p > 1$ **(B)** $p = 0$ or $p \geq 1$ C $p \geq -x$ D $p \geq \sqrt{x}$ ~~E~~ $p \geq 0$ admits $0 < p < 1$ ~~F~~ $p \geq 1$ misses $p = 0$

WHY THIS SHORTCUT IS VALID

The left side is continuous and strictly increasing on $[0, \infty)$ when $p \geq 0$, with range $[\sqrt{p}, \infty)$. This proves both necessity and sufficiency without squaring twice. At $p = 0$, the solution is $x = 0$.

Use the range before doing radical algebra.

TRICK 07 · GUARANTEES AND THE PIGEONHOLE PRINCIPLE

One More Forces It

To find the smallest selection that forces an event, first find the largest selection that can still avoid it.

Largest avoiding selection + 1 = the guarantee.

How to use it

1 Group the combinations that force the event.

For a required sum, pair each number with its partner.

2 Count how many you can safely take.

Take at most one from each dangerous pair. Include any genuinely unpaired numbers.

3 Show the bound can be reached.

Give an avoiding selection of that size. One extra choice must then force the event.

Force a sum of 7

Choose distinct numbers from 1, 2, 3, 4, 5, 6.

The pairs are (1, 6), (2, 5), (3, 4).

Three choices can avoid a pair: take 1, 2, 3. Four choices cannot.

3 safe choices + 1 forces a pair summing to 7.

Small teaching illustration

Watch for this. Watch for numbers with no partner and numbers that would need a second copy of themselves. “Distinct” rules out using the same term twice.

NOW SEE IT IN A TMUA QUESTION

18 How many force a sum of 74?
TMUA 2022 · Paper 2 · Q8



Build the biggest escape, then block it with one more.

How many force a sum of 74?

TMUA 2022 · Paper 2 · Q8

18

FIRST MOVE Build the largest selection avoiding the event, then add one.

Target 30–50 s

From the terms of

$$1, 4, 7, \dots, 70,$$

choose n distinct numbers. What is the smallest n that guarantees two chosen numbers have sum 74?

~~A~~ 12 *not enough*

~~B~~ 13 *still avoidable*

C 14 *13 + 1*

~~D~~ 21 *not smallest*

~~E~~ 22 *not smallest*

~~F~~ 23 *not smallest*

The quick route

✓ Pair the dangerous partners.

$(4, 70), (7, 67), \dots, (34, 40).$

There are 11 pairs.

✓ Keep the two loners.

1 has no partner: 73 is absent.

37 would need another 37, but the choices are distinct.

So 13 numbers can avoid a pair.

✓ One more forces it.

Take at most one from each pair, plus both loners:

$$11 + 2 = 13.$$

The 14th forces 74 → C.

Answer **C**

WHY THIS SHORTCUT IS VALID

The bound 13 is attainable: choose $1, 4, 7, \dots, 37$. Any two distinct numbers in that set sum to at most $34 + 37 = 71$. A 14-element selection must contain both members of one of the 11 pairs.

Count the largest safe selection, then add one.

TRICK 08 · LOGARITHMS WITH CYCLING BASES

Close the Log Loop

When the same three letters rotate through the bases and arguments, change of base gives a product that cancels.

$$\log_a b \times \log_b c \times \log_c a = 1$$

How to use it

1 Spot the closed cycle.

Read the bases and arguments as $a \rightarrow b \rightarrow c \rightarrow a$.

2 Multiply the three logs.

With one common base, each numerator cancels a denominator. The product is 1.

3 Insert the known values.

The remaining log is the reciprocal of the product of the other two.

Two values give the third

Suppose $\log_a b = 2$ and $\log_b c = 3$.

$$2 \times 3 \times \log_c a = 1$$

$$\log_c a = \frac{1}{6}$$

There is no need to find a , b or c .

Small teaching illustration

Watch for this. The logarithms must be defined: bases are positive and not 1, and arguments are positive. Check these conditions before cancelling.

NOW SEE IT IN A TMUA QUESTION

19

Three logs form a loop

TMUA 2022 · Paper 2 · Q15



Close the loop; take the reciprocal.

Three logs form a loop

TMUA 2022 · Paper 2 · Q15

19

FIRST MOVE Multiply around the cycle; the change-of-base fractions cancel.

Target 10–20 s

Real numbers x, y, z are greater than 1 and satisfy

$$\log_x y = z, \quad \log_y z = x.$$

What is $\log_z x$?*The quick route*

↙ Close the cycle $x \rightarrow y \rightarrow z \rightarrow x$.

The product

$\log_x y \log_y z \log_z x$
is 1.

↙ Drop in the two known values.

$$zx \log_z x = 1.$$

$$\log_z x = 1/(xz) \rightarrow \mathbf{F}.$$

↙ Why the loop collapses.

Change every log to natural logs. Each numerator cancels the next denominator.

Answer **(F)**

A y B $\frac{1}{y}$ C xy D $\frac{1}{xy}$ E xz

(F) $\frac{1}{xz}$ *reciprocal of xz*

G yz H $\frac{1}{yz}$

WHY THIS SHORTCUT IS VALID

$\frac{\ln y}{\ln x} \frac{\ln z}{\ln y} \frac{\ln x}{\ln z} = 1$. The assumptions $x, y, z > 1$ ensure all logarithms are defined and all denominators are nonzero. There is no need to solve for x, y or z .

A closed log cycle multiplies to 1.

TRICK 09 · SIGNED AREAS, OVERLAPPING INTEGRALS AND ODDNESS

Cancel Before Calculating

An integral is signed area. Overlap identities and symmetry can determine the answer before you find any antiderivative.

Use cancellation before calculation.

How to use it

1 Check the signs.

Above the axis contributes positively; below contributes negatively. Unsigned area adds both magnitudes.

2 Look for cancellation.

Combine overlapping intervals, or use oddness on an interval symmetric about zero.

3 Return to what was asked.

A zero signed integral can still enclose a large area. Squaring an odd function removes its sign cancellation.

Zero integral, nonzero area

The signed integral over a region is 0. Its only negative part has integral -7 .

The positive part must contribute $+7$.

$$\text{unsigned area} = 7 + 7 = 14$$

The signed total cancels; the two areas add.

Small teaching illustration

Watch for this. *An odd function integrates to zero on a symmetric interval. Its square is non-negative, and has a positive integral when the original function is continuous and not identically zero there.*

NOW SEE IT IN A TMUA QUESTION

20 The quartic you never integrate

TMUA 2018 · Paper 1 · Q12



21 An odd function breaks the claim

TMUA 2016 · Paper 2 · Q18



Signed integral and total area answer different questions.

The quartic you never integrate

TMUA 2018 · Paper 1 · Q12

20

FIRST MOVE Track signed areas or oddness before finding any antiderivative.

Target 30–50 s

Let $f(x) = x(x - p)(x - q)(r - x)$, where $0 < p < q < r$.
Suppose

$$\int_0^r f(x) dx = 0,$$

$$\int_0^q f(x) dx = -2, \quad \int_p^r f(x) dx = -3.$$

Find the total unsigned area between the curve and the axis on $[0, r]$.

The quick route

✓ Read the signs of the factors.

The three lobes have signs

$+, -, +$.

Only the middle one is below the axis.

✓ Isolate the overlap.

Add the two partial integrals; subtract the whole one.

The middle integral is $-2 - 3 - 0 = -5$.

✓ Zero signed total means balance.

Negative magnitude = 5.

Positive area = 5 too.

Total area = 10 → F.

Answer (F)

~~A~~ 0 signed total

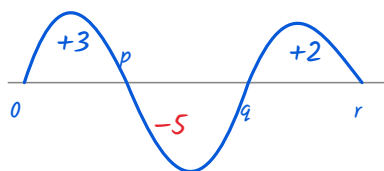
B 1

C 4

~~D~~ 5 only one half

E 6

(F) 10



Teaching sketch · not to scale

WHY THIS SHORTCUT IS VALID

$\int_p^q f = \int_0^q f + \int_p^r f - \int_0^r f$. Since the signed total is zero, positive and negative magnitudes are equal. The individual unsigned lobe areas are 3, 5 and 2, respectively.

Overlapping integrals can be added like overlapping intervals.

An odd function breaks the claim

TMUA 2016 · Paper 2 · Q18

21

FIRST MOVE Track signed areas or oddness before finding any antiderivative.

Target 30–50 s

A claim states that if $f(x)^2 \leq 1$ for all $x \in [-1, 1]$, then

$$\int_{-1}^1 f(x)^2 dx \leq \int_{-1}^1 f(x) dx.$$

Which function is a counterexample?

~~A~~ $f(x) = x + \frac{1}{2}$ *premise fails*

~~B~~ $f(x) = x - \frac{1}{2}$ *premise fails*

~~C~~ $f(x) = x + x^3$ *premise fails*

D $f(x) = x - x^3$

~~E~~ $f(x) = x^2 + x^4$ *premise fails*

~~F~~ $f(x) = x^2 - x^4$ *inequality holds*

The quick route

✓ Check the hypothesis first.

A, B, C and E have $|f| > 1$ at an endpoint. They are not valid counterexamples.

✓ Oddness cancels f.

For $f = x - x^3$:

$$\int_{-1}^1 f = 0.$$

But f^2 is non-negative and not identically zero, so

$$\int_{-1}^1 f^2 > 0.$$

✓ D also meets the condition.

$$|x - x^3| = |x|(1 - x^2) \leq 1 \text{ on the interval.}$$

The condition holds, but the conclusion fails.

D

Answer **D**

WHY THIS SHORTCUT IS VALID

Both factors in $|x|(1 - x^2)$ lie in $[0, 1]$. The function is continuous and nonzero somewhere, so its square has strictly positive integral. F cannot disprove the claim because $0 \leq f \leq 1$ implies $f^2 \leq f$ pointwise.

Oddness cancels the integral; squaring removes the cancellation.

TRICK 10 · EXACTLY-ONE-TRUE LOGIC QUESTIONS

Make the Answers Fight

The options may reveal the answer even when the function or object in the question is unspecified.

An answer cannot be uniquely true if it forces another.

How to use it

1 Translate the wording.

“P only if Q” means $P \rightarrow Q$. “P if Q” means $Q \rightarrow P$.

2 Pair equivalent statements.

A statement and its contrapositive have the same truth value. Neither can be the only true option.

3 Check one-way implications.

If option C being true would also make D true, then C cannot be uniquely true.

Let the choices eliminate each other

Suppose exactly one of A, B, C and D is true.

If A and B are equivalent, eliminate both.

If C implies D, eliminate C.

The exactly-one condition leaves D.

Small teaching illustration

Watch for this. A contrapositive is equivalent to the original statement; its converse need not be. Compare the complete claims, including words such as “every”.

NOW SEE IT IN A TMUA QUESTION

22 Exactly one statement can survive

TMUA 2020 · Paper 2 · Q20



Before solving the object, inspect the relationships between answers.

Exactly one statement can survive

TMUA 2020 · Paper 2 · Q20

22

FIRST MOVE Pair equivalent answers; rule out any option that forces another to be true.

Target 30–50 s

Each statement below is a claim about every real x . Exactly one is true for a function f .

Which statement is it?

~~A~~ $x \geq 0$ only if $f(x) < 0$. *same as B*

~~B~~ $x < 0$ if $f(x) \geq 0$. *same as A*

C $x \geq 0$ only if $f(x) \geq 0$.

~~D~~ $f(x) < 0$ if $x < 0$. *same as E*

~~E~~ $f(x) \geq 0$ only if $x \geq 0$. *same as D*

~~F~~ $f(x) \geq 0$ if and only if $x < 0$. *implies B*

The quick route

✓ Pair equivalent answers.

A and B are contrapositives.
So are D and E .
Each pair has the same truth value.

✓ A pair cannot be uniquely true.

Cross out A , B , D and E .
 F implies B , so F cannot be the only true statement either.

✓ One survivor.

C

“ P only if Q ” means $P \rightarrow Q$.

“ P if Q ” means $Q \rightarrow P$.

Answer **C**

WHY THIS SHORTCUT IS VALID

Equivalent claims cannot have different truth values. A statement that implies another offered statement cannot be uniquely true either. The premise is consistent: $f(x) = 1$ for every x makes C true and all five other universal claims false.

Use the relationships between answers before analysing the unknown function.

The answer — and the move.

The source links open the official paper at the question. Each example page also links to the official answer key.

NO.	OFFICIAL SOURCE	SHORTCUT	KEY
01	TMUA early specimen · Paper 1 · Q7	Small n, Big Cut	C
02	TMUA 2020 · Paper 2 · Q15	Small n, Big Cut	D
03	TMUA 2018 · Paper 2 · Q16	Small n, Big Cut	F
04	TMUA 2019 · Paper 1 · Q4	Shrink the Giant	C
05	TMUA 2017 · Paper 2 · Q6	Shrink the Giant	A
06	TMUA 2023 · Paper 2 · Q17	Shrink the Giant	F
07	TMUA 2017 · Paper 1 · Q19	Choose Your Own Numbers	D
08	TMUA 2023 · Paper 1 · Q11	Choose Your Own Numbers	B
09	TMUA 2023 · Paper 2 · Q19	Choose Your Own Numbers	H
10	TMUA 2023 · Paper 1 · Q10	Follow the π	B
11	TMUA 2022 · Paper 1 · Q19	Follow the π	F
12	TMUA 2021 · Paper 2 · Q7	Mirror, Move, Match	B
13	TMUA 2021 · Paper 2 · Q15	Mirror, Move, Match	B
14	TMUA 2021 · Paper 1 · Q18	Mirror, Move, Match	B
15	TMUA 2022 · Paper 1 · Q15	Beat the Options	H
16	TMUA 2019 · Paper 2 · Q18	Beat the Options	E
17	TMUA 2018 · Paper 2 · Q20	Beat the Options	B
18	TMUA 2022 · Paper 2 · Q8	One More Forces It	C
19	TMUA 2022 · Paper 2 · Q15	Close the Log Loop	F
20	TMUA 2018 · Paper 1 · Q12	Cancel Before Calculating	F
21	TMUA 2016 · Paper 2 · Q18	Cancel Before Calculating	D
22	TMUA 2020 · Paper 2 · Q20	Make the Answers Fight	C

Sources. The questions come from the official UAT UK archive. The early specimen is labelled separately. Wording is rephrased; mathematical conditions and option order are preserved. In example 13, R abbreviates the square-root expression in the original choices. The unnumbered introduction demonstrations are teaching illustrations.

Spot the cue. Make one decisive move.